

BEGINNER LEVEL WORKSHEET: PERCENTAGE

Coverage: Exercises 9A, 9B, 9C, 9D, Mental Maths, Case Studies & HOTS

Total Questions: 40 — **Section A:** 20 MCQs (4 HOTS) — **Section B:** 20 Word Problems**Instructions:** Read each question carefully. Hints are provided to assist your step-by-step problem-solving.**SECTION A: MULTIPLE CHOICE QUESTIONS (20 Questions)***Select the correct option from the given choices for each question.***Q1.** Express $\frac{3}{5}$ as a percentage. (a) 30% (b) 50% (c) 60% (d) 75%**Hint:** To convert any fraction into a percentage, multiply the fraction by 100%.**Q2.** Express 0.06 as a percentage. (a) 0.6% (b) 6% (c) 60% (d) 600%**Hint:** Multiply the decimal number by 100% (move the decimal point two places to the right).**Q3.** What is 2.5% expressed as a fraction in its lowest terms? (a) $\frac{1}{25}$ (b) $\frac{1}{40}$ (c) $\frac{1}{4}$ (d) $\frac{25}{100}$ **Hint:** Remove the % sign and divide by 100: $\frac{2.5}{100} = \frac{25}{1000}$. Simplify.**Q4.** Find 15% of Rs 240. (a) Rs 24 (b) Rs 30 (c) Rs 36 (d) Rs 40**Q5.** What per cent of 2 kg is 250 g? (a) 12.5% (b) 25% (c) 10% (d) 20%**Hint:** Convert both quantities to the same unit first: 2 kg = 2000 g. Required % = $\frac{250}{2000} \times 100\%$.**Q6.** If 20% of a number is 45, find the number. (a) 180 (b) 200 (c) 225 (d) 250**Hint:** Let the number be x . Set up the equation $\frac{20}{100} \times x = 45$.**Q7.** In a class of 50 pupils, 8% remained absent on a rainy day. How many pupils were present? (a) 42 (b) 44 (c) 46 (d) 48**Q8.** Increase 150 by 20%. (a) 170 (b) 180 (c) 190 (d) 200**Hint:** Increase = 20% of 150 = 30. New value = 150 + 30.**Q9.** Decrease 200 by 15%. (a) 170 (b) 175 (c) 180 (d) 185**Q10.** By what decimal multiplier must a given number be multiplied to increase it by 25%? (a) 0.25 (b) 1.25 (c) 2.5 (d) 125**Hint:** Increased value = $x + 0.25x = 1.25x$.**Q11.** What per cent of a day (24 hours) is 3 hours? (a) 8% (b) 12.5% (c) 15% (d) 25%

Q12. Assertion (A): 1% of 100 is 1, and 100% of 1 is also 1.

Reason (R): $x\%$ of $y = y\%$ of x .

- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).
- (c) (A) is true but (R) is false.
- (d) (A) is false but (R) is true.

Q13. In a grid of 20 equal squares, 7 squares are shaded. What percentage of the figure is shaded?

- (a) 28% (b) 35% (c) 40% (d) 70%

Q14. Case Study: A mixture contains 200 g of Salt A and 300 g of Salt B. What is the percentage of Salt A in the mixture? (a) 20% (b) 40% (c) 50% (d) 60%

Hint: Total mass of mixture = 200 g + 300 g = 500 g. Percentage = $\frac{200}{500} \times 100\%$.

Q15. Ameena has 10 coins of Rs 1 and 10 coins of Rs 5. What percentage of the **total value** of coins is in Rs 5 coins? (a) 50% (b) 60% (c) 75% (d) $83\frac{1}{3}\%$

Hint: Total value = $(10 \times 1) + (10 \times 5) = \text{Rs } 60$. Value of Rs 5 coins = Rs 50.

Q16. A sum of money is divided between A and B in the ratio 3 : 2. What is A's share expressed as a percentage of the total sum? (a) 30% (b) 40% (c) 60% (d) 75%

Q17. [HOTS] The price of an item is first increased by 20% and then the new price is decreased by 20%. What is the net percentage change in the price? (a) No change (b) 4% increase (c) 4% decrease (d) 2% decrease

Hint: Assume original price is Rs 100. After 20% increase, price = Rs 120. Now decrease 120 by 20% (20% of 120 = 24). Final price = $120 - 24 = 96$.

Q18. [HOTS] In a competitive examination, 40% of candidates failed in Mathematics and 30% failed in English. If 10% failed in both subjects, what percentage of candidates passed in both subjects? (a) 30% (b) 40% (c) 50% (d) 60%

Hint: Total percentage failed = (% failed Math) + (% failed English) – (% failed both) = $40 + 30 - 10 = 60\%$.

Q19. [HOTS] In an election between two candidates, 10% of total votes polled were declared invalid. The winning candidate secured 60% of the valid votes and won by 1,800 votes. What was the total number of votes polled? (a) 9,000 (b) 10,000 (c) 12,000 (d) 15,000

Hint: Let valid votes be V . Winner gets 60%, loser gets 40%. Difference = 20% of $V = 1800 \Rightarrow V = 9000$. Since 10% votes were invalid, 90% of total votes = 9000.

Q20. [HOTS] A person spends 75% of his income. If his income increases by 20% and his expenditure increases by 10%, by what percentage do his savings increase? (a) 25% (b) 30% (c) 40% (d) 50%

Hint: Let original income = 100, expenditure = 75, savings = 25. New income = 120, new expenditure = $75 \times 1.10 = 82.5$. New savings = $120 - 82.5 = 37.5$.

SECTION B: STATEMENT BASED & WORD PROBLEMS (20 Questions)

Solve each problem step-by-step. Show all intermediate working.

Q21. Basic Conversions to Percentage:

- (a) Convert $\frac{7}{20}$ into a percentage.
- (b) Convert 0.08 into a percentage.
- (c) Convert the ratio 4 : 25 into a percentage.

Q22. Converting Percentages to Simplified Forms:

- (a) Express 45% as a fraction in its lowest terms.
- (b) Express 18.5% as a decimal.

Q23. Find 12.5% of Rs 1,600.

Hint: $12.5\% = \frac{12.5}{100} = \frac{1}{8}$. Calculate $\frac{1}{8} \times 1600$.

Q24. Express 45 cm as a percentage of 3 metres.

Hint: Convert 3 m to centimeters (3 m = 300 cm). Required percentage = $\frac{45}{300} \times 100\%$.

Q25. If 16% of a number is 72, find the number.**Q26. A school football team played 25 matches in a season and won 15 of them. What per cent of the matches did they win, and what per cent did they lose?****Q27. In a school of 1,200 students, 15% were absent on a rainy day. How many students were present on that day?****Q28. Mr. Verma earns Rs 40,000 per month. He spends 82% of his income and saves the rest. How much money does he save each month?**

Hint: Percentage of income saved = $100\% - 82\% = 18\%$. Calculate 18% of Rs 40,000.

Q29. Rohit scored 280 marks out of a maximum of 400 marks in an examination. Calculate his percentage score.**Q30. An alloy consists of 13 parts of copper, 7 parts of zinc, and 5 parts of nickel. Find the percentage of copper in the alloy.**

Hint: Total parts in alloy = $13 + 7 + 5 = 25$. Percentage of copper = $\frac{13}{25} \times 100\%$.

Q31. Increase Rs 450 by 12%.**Q32. The price of a shirt was reduced from Rs 800 to Rs 680. Find the percentage decrease in its price.**

Hint: Decrease in price = $800 - 680 = \text{Rs } 120$. Percentage decrease = $\frac{\text{Decrease}}{\text{Original Price}} \times 100\%$.

Q33. After deducting 5% tax from a telephone bill, the remaining amount to be paid is Rs 1,140. Find the original bill amount.

Hint: Let original bill be x . Remaining percentage = $100\% - 5\% = 95\%$. So, 95% of $x = 1140$.

Q34. Multiplier Rules:

- (a) By what decimal number should a given quantity be multiplied to increase it by 15%?
- (b) By what decimal number should a given quantity be multiplied to decrease it by 20%?

Q35. A car was purchased for Rs 5,00,000. Its value depreciated by 10% after one year. What is the value of the car now?

Q36. A line segment of length 8 cm was wrongly measured as 8.2 cm. Calculate the error percentage.

Hint: Error = $8.2 - 8.0 = 0.2$ cm. Error percentage = $\frac{\text{Error}}{\text{Actual Length}} \times 100\%$.

Q37. Mental Maths & Conceptual Verification:

- (a) Is 20% of 50 equal to 50% of 20? Explain briefly.
- (b) Fill in the blank: 15% of 80 is _____.

Q38. Case Study (Chemical Mixture): A chemist prepares Mixture X by combining 150 g of Chemical A, 250 g of Chemical B, and 100 g of Chemical C.

- (a) Find the percentage of Chemical A in the mixture.
- (b) Find the percentage of Chemical B in the mixture.

Q39. Competency Focused Question: A sum of Rs 12,000 is divided among three friends A, B, and C in the ratio 2 : 3 : 5. Express each person's share as a percentage of the total sum and calculate their respective share amounts.

Q40. [HOTS Word Problem] Mrs. Anita earns Rs 50,000 per month. 10% of her income is deducted for her Provident Fund (PF). From the remaining income, she spends 30% on house rent, 40% on food, and 10% on entertainment, saving the rest.

- (a) How much money is credited each month to her Provident Fund?
- (b) How much money does she spend on food?
- (c) What is her total monthly savings in Rupees?

Hint: First compute PF = 10% of 50,000 = 5,000. Remaining income = Rs 45,000. Spending percentages apply to this remaining income of Rs 45,000.

SECTION C: ANSWER KEY & SOLUTIONS

Q#	Correct Answer & Brief Explanation
1	(c) 60% ($\frac{3}{5} \times 100\% = 60\%$)
2	(b) 6% ($0.06 \times 100\% = 6\%$)
3	(b) 1/40 ($\frac{2.5}{100} = \frac{25}{1000} = \frac{1}{40}$)
4	(c) Rs 36 ($0.15 \times 240 = 36$)
5	(a) 12.5% ($\frac{250}{2000} \times 100\% = 12.5\%$)
6	(c) 225 ($x = 45/0.20 = 225$)
7	(c) 46 ($100\% - 8\% = 92\%$; $0.92 \times 50 = 46$)
8	(b) 180 ($150 + 0.20 \times 150 = 150 + 30 = 180$)
9	(a) 170 ($200 - 0.15 \times 200 = 200 - 30 = 170$)
10	(b) 1.25 ($1 + 0.25 = 1.25$)
11	(b) 12.5% ($\frac{3}{24} \times 100\% = \frac{1}{8} \times 100\% = 12.5\%$)
12	(a) Both (A) and (R) are true, and (R) is correct explanation
13	(b) 35% ($\frac{7}{20} \times 100\% = 35\%$)
14	(b) 40% ($\frac{200}{500} \times 100\% = 40\%$)
15	(d) 83 1/3% (Total value = Rs 60; Rs 5 coins value = Rs 50; $\frac{50}{60} \times 100\% = 83\frac{1}{3}\%$)
16	(c) 60% ($\frac{3}{3+2} \times 100\% = \frac{3}{5} \times 100\% = 60\%$)
17	(c) 4% decrease ($100 \rightarrow 120 \rightarrow 120 - 24 = 96$; Net decrease = 4%)
18	(b) 40% (Total failed = $40 + 30 - 10 = 60\%$; Passed = $100 - 60 = 40\%$)
19	(b) 10,000 (20% of valid = 1800 $\implies$ Valid = 9000; Total = $9000/0.9 = 10000$)
20	(d) 50% (Orig: Inc 100, Exp 75, Sav 25; New: Inc 120, Exp 82.5, Sav 37.5; % Inc = $\frac{12.5}{25} \times 100\% = 50\%$)

Section B Answers:

Q21. (a) 35%, (b) 8%, (c) 16%

Q22. (a) $\frac{9}{20}$, (b) 0.185

Q23. Rs 200

Q24. 15%

Q25. 450

Q26. Won = 60%, Lost = 40%

Q27. 1,020 students present

Q28. Rs 7,200 saved per month

Q29. 70%

Q30. 52% copper

Q31. Rs 504

Q32. 15% decrease

Q33. Rs 1,200

Q34. (a) 1.15, (b) 0.80

Q35. Rs 4,50,000

Q36. 2.5% error

Q37. (a) Yes, both equal 10. (20% of $50 = 10$ and 50% of $20 = 10$), (b) 12

Q38. (a) 30%, (b) 50%

Q39. A's share = 20% (Rs 2,400), B's share = 30% (Rs 3,600), C's share = 50% (Rs 6,000)

Q40. (a) PF = Rs 5,000, (b) Food = Rs 18,000, (c) Savings = Rs 9,000